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Green energy of discrete signed measure on concentric circles

V. N. Dubinin

Abstract. We show that the difference between the Green energy of a discrete signed measure relative to a circular annulus concentrated at some points on concentric circles and the energy of the signed measure at symmetric points is non-decreasing during the expansion of the annulus. As a corollary, generalizations of the classical Pólya–Schur inequality for complex numbers are obtained. Some open problems are formulated.

Keywords: Green function, Green energy, capacity of condensers, dissymmetrization, inequality.

§ 1. Introduction and statement of the main result

The symmetry of an extremal object has always aroused the interest. However, in a number of cases, it does not seem possible to find such a symmetry, even for problems in simple form. The foregoing applies equally well to the extremal problem on the energy of a discrete measure concentrated at some equal distance from a given point. The involved problems for various kinds of energies of discrete measures, both on the plane, and in spaces of high dimension, have been extensively studied (see, for example, [1]–[6] and the references cited there). Less attention has been paid to the extremal properties of the Green energy, which is the energy generated by the Green kernel $g_B(z,\zeta)$ (see [7]). Here, $g_B(z,\zeta)$ is the classical Green function of a domain $B \subset \overline{\mathbb{C}}$, which is zero outside B. The interest in the behaviour of the Green energy stems from applications in geometric theory of holomorphic functions (see [8]). We mention the following natural problems in this regard. For which domains B is the Green energy of a discrete measure concentrated at symmetric points minimal? How does the difference between the energy of an arbitrary and the extremal measures vary with B? What can be said about the energy of a discrete signed measure concentrated at points on some concentric circles? What are the extremal properties of the mutual energy of signed measures? Partial answers to these problems were given in [9] and [10]. Below, we will provide the proof of a strengthened version of Theorem 1, which was announced in [10]. Consider the following configuration. We fix a natural number $m \geqslant 1$ and

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non-negative numbers

$$0 \leqslant s_1 \leqslant t_1 < \rho_1 < \rho_2 < \cdots < \rho_m < t_2 \leqslant s_2 \leqslant \infty.$$

For arbitrary real numbers θ_j, $j = 1, \ldots, n$, $n \geqslant 2$, satisfying

$$\theta_1 < \theta_2 < \cdots < \theta_n < \theta_1 + 2\pi,$$

we let $Z = \{z_k\}_{k=1}^{mn}$ denote the set of all possible points on $\mathbb{C}$ which are the points of intersection of the circles $|z| = \rho_k$, $k = 1, \ldots, m$, with the rays $\arg z = \theta_j$, $j = 1, \ldots, n$. Let $\Delta = \{\delta_k\}_{k=1}^{mn}$ be an arbitrary discrete signed measure equal to δ_k at the points z_k and such that $\delta_k = \delta_{k'}$ for $|z_k| = |z_{k'}|$, $1 \leqslant k, k' \leqslant mn$. The Green energy of this signed measure relative to the circular annulus $B(s_1, s_2) := \{z\colon s_1 < |z| < s_2\}$ is denoted by

$$E(Z, \Delta, B(s_1, s_2)) = \sum_{k=1}^{mn} \sum_{\substack{l=1 \\ l \neq k}}^{mn} \delta_k \delta_l g_{B(s_1, s_2)}(z_k, z_l).$$

The main result of the present paper is the following

Theorem 1.1. *In the above notation,*

$$E(Z, \Delta, B(s_1, s_2)) - E(Z^*, \Delta, B(s_1, s_2)) \geqslant E(Z, \Delta, B(t_1, t_2)) - E(Z^*, \Delta, B(t_1, t_2)), \tag{1.1}$$

where different symmetric points from $Z^* = \{z_k^*\}_{k=1}^{mn}$ *are given by* $|z_k^*| = |z_k|$, $\arg (z_k^*)^n = 0$, $k = 1, \ldots, mn$.

From Theorem 4.15 in [8] it follows that both differences in inequality (1.1) are non-negative. Thus, we find the variation of the departure of the Green energy from the extremal one during the expansion of the annulus $B(t_1, t_2)$. In addition, we will consider signed measures that assume values of different signs and are concentrated on several circles, rather than on a single circle (cf. [9]). It should be noted that inequality (1.1) is non-trivial — this is seen, for example, from the analytic representation of the Green function of the circular annulus in terms of theta-functions (see [11], § 55). The proof of Theorem 1.1, which is given below in § 3, is based on a geometric dissymmetrization type transformation (see [8], § 4.4) and an asymptotic formula for the capacity of a generalized condenser (see [8], § 2.2, [12], and § 2 below). Our approach is also capable of providing an analogue of Theorem 1.1 in the setting described in Theorem 2 of [10]. Setting $s_1 = 0$ or $t_1 = 0$ in (1.1), we arrive at inequalities for the difference of the discrete Green energies relative to a disc and an annulus, or relative to a disc and a different disc. In § 4, we obtain from these inequalities some relations for complex numbers. Using the energetic interpretation of the resulting inequalities, we can, in particular, compare the mutual logarithmic energy of discrete measures concentrated on two concentric circles with the mutual energy of symmetric measures, and also compare the mutual Green energies relative to a disc. Some open problems are given in § 5. Some results of the present paper were discussed on November 12, 2018, at the seminar on complex analysis at the Mathematical Institute of the Russian Academy of Sciences (Gonchar's seminar).

§ 2. Auxiliary results

In what follows, (γ, Γ) denotes a doubly connected domain on the plane $\mathbb{C}$ bounded by closed curves γ and Γ. We will assume that the curves γ and Γ are analytic and are traversed positively relative to the domain (γ, Γ). Let $\varphi(z)$ be a real-valued continuously differentiable function on γ. For an arbitrary sufficiently small $\varepsilon > 0$, consider the "deformation" of the curve γ defined by

$$\delta n(z) := \varepsilon\varphi(z) + O(\varepsilon^2). \tag{2.1}$$

Under this deformation, γ is transformed to the curve

$$\gamma' = \left\{ z' = z + \frac{\delta n(z) i\, dz}{|dz|},\ z \in \gamma \right\}.$$

Here, $\delta n(z)$ is a continuously differentiable function on γ, and $O(\varepsilon^2)$ is estimated uniformly on γ.

The next result can be looked upon as a particular case of the Hadamard variational formula for Dirichlet integrals of harmonic measures [13], A3.11, which, in turn, follows from the variational formula for Green functions (see [13], A3.3, and [14]).

Lemma 2.1. *Under the above assumptions, the moduli of doubly connected domains behave asymptotically, as* $\varepsilon \to 0$,

$$\operatorname{mod}(\gamma', \Gamma) = \operatorname{mod}(\gamma, \Gamma) - (\operatorname{mod}(\gamma, \Gamma))^2 \int_\gamma \left(\frac{\partial \omega}{\partial n} \right)^2 \delta n(z)\, |dz| + O((\delta n)^2), \tag{2.2}$$

where $\omega(z)$ *is the harmonic measure of the curve* γ *relative to the domain* (γ, Γ) *evaluated at* z, *and* $\delta n = \max\{|\delta n(z)| \colon z \in \gamma\}$.

We set

$$I(v, G) = \iint_G |\nabla v|^2\, dx\, dy. \tag{2.3}$$

The next result is a consequence of the asymptotic formula for capacities of condensers (see [12], Theorem 1).

Lemma 2.2. *Let a domain* D *admit the classical Green function, let* $\{\zeta_k\}_{k=1}^N$ *be a set of different finite points in* D, *and let*

$$\delta_k(r) = \delta_k + \frac{c_k}{\log r} + O\left(\left(\frac{1}{\log r} \right)^2 \right), \qquad r \to 0,$$

where $\delta_k \neq 0$ *and* c_k *are real constants,* $k = 1, \dots, N$. *Net, let closed sets* $\mathscr{E}(\zeta_k, r)$ *be bounded by smooth Jordan curves on which*

$$|z - \zeta_k| = r\left(1 + O\left(\frac{1}{\log r} \right) \right), \qquad r \to 0, \quad k = 1, \dots, N.$$

Also let a real-valued function u be continuous in $\overline{D}$, harmonic in $D\setminus\bigcup_{k=1}^{N}\mathscr{E}(\zeta_k,r)$, vanish on ∂D, and $\delta_k(r)$ on $\mathscr{E}(\zeta_k,r)$, $k=1,\dots,N$. Then

$$I\biggl(u, D\setminus\bigcup_{k=1}^{N}\mathscr{E}(\zeta_k,r)\biggr)=-2\pi\sum_{k=1}^{N}\frac{\delta_k^2}{\log r}-2\pi\biggl\{\sum_{k=1}^{N}[2c_k\delta_k+\delta_k^2\log r(D,\zeta_k)]+\sum_{k=1}^{N}\sum_{\substack{l=1\\ l\neq k}}^{N}\delta_k\delta_l g_D(\zeta_k,\zeta_l)\biggr\}\biggl(\frac{1}{\log r}\biggr)^2+o\biggl(\biggl(\frac{1}{\log r}\biggr)^2\biggr),\qquad r\to 0,$$

where $r(D,\zeta_k)$ is the inner radius of the domain D relative to the point ζ_k, $k=1,\dots,N$ (see [8], *§ 2.1).*

Numerous applications of the dissymmetrization , which we describe below, can be found in the book [8]. Let $n\geqslant 2$ be a natural number, and let

$$L_j^*=\biggl\{z\colon\ \arg z=\frac{2\pi j}{n}\biggr\},\qquad j=1,\dots,n.$$

Next, let Φ be the symmetry group on $\overline{\mathbb{C}}$ consisting of the superpositions of the reflections about the straight lines passing through the rays L_j^*, $j=1,\dots,n$, and also about the straight lines passing through the bisectors of the angles formed by these rays. It is clear that, for odd n, the last requirement can be omitted. The group Φ is the "dihedral group". Throughout the paper, the symmetry will mean invariance under mappings of the group Φ. A set $A\subset\overline{\mathbb{C}}$ will be said to be *symmetric* (relative to the group Φ) if $\varphi(A)=A$ for any isometry $\varphi\in\Phi$. A real-valued function v defined on a symmetric set Ω is called symmetric if it satisfies $v(z)\equiv v(\varphi(z))$ for any $\varphi\in\Phi$. A set of closed angles with vertices at the origin will be called a *decomposition* of the complex sphere $\overline{\mathbb{C}}$ if any two angles from this set have no common interior points, and the union of all angles coincides with $\overline{\mathbb{C}}$. Let $\{P_j\}_{j=1}^{j_0}$ be a symmetric decomposition of $\overline{\mathbb{C}}$, that is, $\{\varphi(P_j)\}_{j=1}^{j_0}=\{P_j\}_{j=1}^{j_0}$ for any isometry $\varphi\in\Phi$. A set of rotations $\{\lambda_j\}_{j=1}^{j_0}$ of the form $\lambda_j(z)=e^{i\varphi_j}z$, $j=1,\dots,j_0$, will be called a *dissymmetrization* of a symmetric decomposition $\{P_j\}_{j=1}^{j_0}$ if the set of ranges $\{S_j\}_{j=1}^{j_0}$, $S_j=\lambda_j(P_j)$, $j=1,\dots,j_0$, is also a decomposition of $\overline{\mathbb{C}}$ and if the following condition is met:

$(*)$ for any nonempty intersection $S_j\cap S_{j'}$, $1\leqslant j,j'\leqslant j_0$, there exists an isometry $\varphi\in\Phi$ such that $\varphi(\lambda_j^{-1}(S_j\cap S_{j'}))=\lambda_{j'}^{-1}(S_j\cap S_{j'})$.

Let A be an arbitrary subset of the sphere $\overline{\mathbb{C}}$ and v be a symmetric function defined on a symmetric set Ω. We set

$$\operatorname{Dis}A=\bigcup_{j=1}^{j_0}\lambda_j(A\cap P_j),$$

$$\operatorname{Dis}v(z)=v(\lambda_j^{-1}(z)),\qquad z\in S_j\cap\operatorname{Dis}\Omega,\quad j=1,\dots,j_0.$$

By symmetry of the function v and in view of condition $(*)$, the function $\operatorname{Dis}v$ is uniquely defined on $\operatorname{Dis}\Omega$. We say that a set A (respectively, a function v) is transformed to the set $\operatorname{Dis}A$ (the function $\operatorname{Dis}v$) under dissymmetrization $\{\lambda_j\}_{j=1}^{j_0}$.

Lemma 2.3 (see [8], Lemma 4.2). *Let θ_j, $j = 1, \dots, n$, $n \geqslant 2$, be arbitrary numbers from Theorem 1.1, let*

$$L_j = \{z\colon \arg z = \theta_j\}, \qquad j = 1, \dots, n,$$

and let ψ be the smallest positive angle between the rays L_j. Then there exist a symmetric decomposition $\{P_j\}_{j=1}^{j_0}$, $j_0 \geqslant n$, and a dissymmetrization $\{\lambda_j\}_{j=1}^{j_0}$ such that each ray L_j^* is the bisector of the angle P_j of size ψ, and $\operatorname{Dis} L_j^* = L_j$, $j = 1, \dots, n$.

§ 3. Proof of Theorem 1.1

It can be assumed that $0 < s_1 < t_1$, $t_2 = s_2 = 1$, and $\delta_k \neq 0$, $k = 1, \dots, mn$. Consider the difference of the energies of the signed measures on the left of (1.1) in the annulus $B(s, 1)$ ($s_1 = s$, $s_2 = 1$) as a function of parameter s, $0 \leqslant s < \rho_1$. It suffices to show that the derivative of this function is non-positive on $(0, \rho_1)$. We fix s and Δs, $0 < s < s + \Delta s < s + 2\Delta s < \rho_1$, and define[1]

$$t = s + \Delta s, \qquad \tau = s + 2\Delta s, \qquad T(R) \equiv T_z(R) = \{z\colon |z| = R\}, \qquad T = T(1);$$

$$g_s^*(z) = \sum_{k=1}^{mn} \delta_k \sum_{l=1}^{mn} \beta_{lk} g(z, z_k^*), \qquad z \in B(s, 1) = (T(s), T),$$

where

$$\beta_{lk} = \begin{cases} -\dfrac{g(z_l^*, z_k^*)}{(\log r)^2}, & l \neq k, \\ -\dfrac{1}{\log r}\left[1 + \dfrac{\log r((T(s), T), z_k^*)}{\log r}\right], & l = k, \end{cases}$$

$g(z, \zeta)$ is the Green function of the domain $(T(s), T)$, $r((T(s), T), z)$ is the inner radius of this domain relative to the point z, and $r > 0$ is sufficiently small;

$$\mathscr{E}(z_k^*, r) = \left\{z\colon \frac{g_s^*(z)}{\delta_k} \geqslant 1\right\}, \qquad k = 1, \dots, mn,$$

$$\mathscr{E}^*(r) = \bigcup_{k=1}^{mn} \mathscr{E}(z_k^*, r) \subset (T(s), T).$$

Let $b(z) := \log |z|$, and let

$$u^* := b + g_s^*.$$

Our next aim is to construct a permutation in the range of values of the argument of the function u^* under which the new function v^* (3.7) has the same Dirichlet integral as u^*, the level set $v^* = b(t)$ is a circle centred at the origin, and $v^* \equiv u^*$ near the set $\mathscr{E}^*(r)$. By the definition of the function g_s^*, the (Hausdorff) distance between the curve

$$\gamma_t^*\colon u^* = b(t)$$

[1] Here and in what follows, we will also use the notation from § 2.

and the circle $T(t)$ behaves as[2] $O'(\Delta s/\log r)$. Moreover, as $r \to 0$, the normals to the curves γ_t^* and $T(t)$ at the corresponding points come closer to each other, so that the deformation $\delta n(z)$ of the circle $T(t)$ to the curve γ_t^* has the form (2.1), where $\varepsilon = -1/\log r$ and $\delta n = O'(\Delta s/\log r)$. In what follows, we will deal only with such deformations of the curves (without special mention). The above pertains to the curve $\gamma_\tau^*\colon u^* = b(\tau)$ and the circle $T(\tau)$. The curve γ_t^* is a level line of the harmonic measure γ_τ^* relative to the domain $(T(s), \gamma_\tau^*)$. By Grotzsch's lemma (see, for example, [8], Theorem 1.14),

$$\operatorname{mod}(T(s), \gamma_t^*) + \operatorname{mod}(\gamma_t^*, \gamma_\tau^*) = \operatorname{mod}(T(s), \gamma_\tau^*). \tag{3.1}$$

Similarly, we have

$$\operatorname{mod}(T(s), T(t)) + \operatorname{mod}(T(t), T(\tau)) = \operatorname{mod}(T(s), T(\tau)). \tag{3.2}$$

By Lemma 2.1,

$$\operatorname{mod}(T(s), \gamma_t^*) = \operatorname{mod}(T(s), T(t)) - \big(\operatorname{mod}(T(s), T(t))\big)^2 \int_{T(t)} \left(\frac{\partial \omega}{\partial n}\right)^2 \delta n(z)\,|d(z)| + O'\left(\left(\frac{\Delta s}{\log r}\right)^2\right),$$

where

$$\operatorname{mod}(T(s), T(t)) = \frac{1}{2\pi}\log\frac{t}{s}, \qquad \omega(z) = \frac{\log(|z|/s)}{\log(t/s)}, \qquad \delta n(z) = O'\left(\frac{\Delta s}{\log r}\right).$$

Hence

$$\operatorname{mod}(T(s), \gamma_t^*) = \operatorname{mod}(T(s), T(t)) + O'\left(\frac{\Delta s}{\log r}\right). \tag{3.3}$$

Similarly, we have

$$\operatorname{mod}(T(s), \gamma_\tau^*) = \operatorname{mod}(T(s), T(\tau)) + O'\left(\frac{\Delta s}{\log r}\right).$$

Now an appeal to (3.1), (3.2) and (3.3) shows that

$$\operatorname{mod}(\gamma_t^*, \gamma_\tau^*) = \operatorname{mod}(T(t), T(\tau)) + O'\left(\frac{\Delta s}{\log r}\right). \tag{3.4}$$

By continuity and monotonicity of the modulus, for each r, there exists a value $t(r)$ such that

$$\operatorname{mod}(T(t(r)), \gamma_\tau^*) = \operatorname{mod}(\gamma_t^*, \gamma_\tau^*),$$

and the departure of the circle $T(t(r))$ from the curve γ_t^* (and from $T(t)$) behaves as $O'(\Delta s/\log r)$. In particular, $t(r) = t + O'(\Delta s/\log r)$. In addition, by definition of the curve γ_t^* we have

$$t(r) = t + \frac{c}{\log r} + O\left(\left(\frac{1}{\log r}\right)^2\right), \qquad r \to 0,$$

where c is some constant.

[2]The prime in the Big-Oh notation means that the corresponding quantity admits a uniform estimate for all sufficiently small Δs as $r \to 0$.

Another appeal to Grotzsch's lemma shows that

$$\operatorname{mod}\bigl(T(s),T(t(r))\bigr)+\operatorname{mod}\bigl(T(t(r)),\gamma_\tau^*\bigr)\leqslant\operatorname{mod}(T(s),\gamma_\tau^*). \tag{3.5}$$

Let us apply (3.3) and (3.4) to verify that

$$\operatorname{mod}\bigl(T(s),T(t(r))\bigr)+\operatorname{mod}\bigl(T(t(r)),\gamma_\tau^*\bigr)=\operatorname{mod}(T(s),\gamma_\tau^*)+O'\biggl(\biggl(\frac{\Delta s}{\log r}\biggr)^2\biggr). \tag{3.6}$$

The second modulus in (3.6) cannot be so easily dealt with via the Hadamard formula (2.2) because both the boundary components of the domain $(T(t(r)),\gamma_\tau^*)$ vary as $r\to 0$. So, we will apply the conformal version of (3.6). Let a function $\zeta=F_r(z)$ map conformally and univalently the domain $(T(s),\gamma_\tau^*)$ onto the annulus $R_s(r)<|\zeta|<R_\tau(r)$ so that $F_r(T(s))=T_\zeta(R_s(r))$, $F_r(\gamma_t^*)=T_\zeta(t)$, $F_r(s)>0$. Using (3.3) and (3.4), we get

$$\frac{s}{R_s(r)}=1+O'\biggl(\frac{\Delta s}{\log r}\biggr),\qquad \frac{\tau}{R_\tau(r)}=1+O'\biggl(\frac{\Delta s}{\log r}\biggr).$$

Since $F_r(z)\rightrightarrows z$, $r\to 0$, uniformly in the fixed neighbourhood $T(t)$, the distance between the curves $\Gamma(r):=F_r(T(t(r)))$ and $T_\zeta(t)$ behaves as $O'(\Delta s/\log r)$. Let $\delta n(\zeta)$ be a deformation of the circle $T_\zeta(t)$ which transforms it to the curve $\Gamma(r)$ in the direction of the inward normal vector relative to the disc $|\zeta|<t$ (the deformation in the direction of the outward normal is $-\delta n(\zeta)$). In order to apply the Hadamard formula one should fix both boundary circles of the original domain. So, we consider the additional homothety

$$\frac{s\zeta}{R_s(r)}\colon\bigl(T_\zeta(R_s(r)),T_\zeta(t)\bigr)\to\biggl(T_w(s),T_w\biggl(\frac{st}{R_s(r)}\biggr)\biggr)$$

under which the curve $\Gamma(r)$ is transformed to some curve $\Gamma_s(r)$ obtained from the circle $T_w(st/R_s(r))$ via the deformation $(s/R_s(r))\delta n(R_s(r)w/s)$. Let $\delta n_s(w)=t-st/R_s(r)$ be the deformation that maps the circle $T_w(t)$ to the circle $T_w(st/R_s(r))$ so that the curve $\Gamma_s(r)$ can be considered as the range of $T_w(t)$ under the deformation

$$\delta n_s(w)+\frac{s}{R_s(r)}\,\delta n(w),\qquad w\in T(t).$$

The harmonic measure of the circle $T_w(t)$ relative to the annulus $s<|w|<t$ is

$$\omega(w)=\frac{\log|w/s|}{\log|t/s|}.$$

Hence $(\partial\omega/\partial n)^2 = (t\log(t/s))^{-2}$ on $T_w(t)$, and now, from (2.2) and by conformal invariance of the modulus, we have

$$\begin{aligned}
&\operatorname{mod}\bigl(T(s), T(t(r))\bigr) = \operatorname{mod}\bigl(T_\zeta(R_s(r), \Gamma(r))\bigr) = \operatorname{mod}(T_w(s), \Gamma_s(r)) \\
&= \operatorname{mod}(T_w(s), T_w(t)) \\
&\quad - [\operatorname{mod}(T_w(s), T_w(t))]^2 \int_{T_w(t)} \left(\frac{\partial\omega}{\partial n}\right)^2 \left(\delta n_s(w) + \frac{s\delta n(w)}{R_s(r)}\right) |dw| + O'\left(\left(\frac{\Delta s}{\log r}\right)^2\right) \\
&= \frac{1}{2\pi}\log\frac{t}{s} - \frac{1}{t(2\pi)^2}\int_0^{2\pi}\left(\delta n_s(te^{i\theta}) + \frac{s}{R_s(r)}\,\delta n(te^{i\theta})\right) d\theta + O'\left(\left(\frac{\Delta s}{\log r}\right)^2\right) \\
&= \frac{1}{2\pi}\log\frac{t}{R_s(r)} - \frac{1}{t(2\pi)^2}\int_0^{2\pi}\frac{s}{R_s(r)}\,\delta n(te^{i\theta})\, d\theta + O'\left(\left(\frac{\Delta s}{\log r}\right)^2\right) \\
&= \frac{1}{2\pi}\log\frac{t}{R_s(r)} - \frac{1}{t(2\pi)^2}\int_0^{2\pi}\delta n(te^{i\theta})\, d\theta + O'\left(\left(\frac{\Delta s}{\log r}\right)^2\right).
\end{aligned}$$

Similarly, the homothety

$$\frac{\tau\zeta}{R_\tau(r)} \colon \bigl(T_\zeta(t), T_\zeta(R_\tau(r))\bigr) \to \left(T_w\left(\frac{\tau t}{R_\tau(r)}\right), T_w(\tau)\right)$$

transforms $\Gamma(r)$ to the curve $\Gamma_\tau(r)$, which is obtained from $T_w(t\tau/R_\tau(r))$ by the deformation $-(\tau/R_\tau(r))\delta n(R_\tau(r)w/\tau)$. Let $\delta n_\tau(w)$ send $T_w(t)$ to $T_w(t\tau/R_\tau(r))$, so that the curve $\Gamma_\tau(r)$ is obtained from the circle $T_w(t)$ via the deformation

$$\delta n_\tau(w) - \frac{\tau}{R_\tau(r)}\,\delta n(w).$$

Now using (2.2) we obtain

$$\begin{aligned}
\operatorname{mod}\bigl(T(t(r)), \gamma_\tau^*\bigr) &= \operatorname{mod}\bigl(\Gamma(r), T_\zeta(R_\tau(r))\bigr) = \operatorname{mod}(\Gamma_\tau(r), T_w(\tau)) \\
&= \frac{1}{2\pi}\log\frac{R_\tau(r)}{t} + \frac{1}{t(2\pi)^2}\int_0^{2\pi}\delta n(te^{i\theta})\, d\theta + O'\left(\left(\frac{\Delta s}{\log r}\right)^2\right).
\end{aligned}$$

Hence

$$\begin{aligned}
\operatorname{mod}\bigl(T(s), T(t(r))\bigr) + \operatorname{mod}\bigl(T(t(r)), \gamma_\tau^*\bigr) &= \frac{1}{2\pi}\log\frac{R_\tau(r)}{R_s(r)} + O'\left(\left(\frac{\Delta s}{\log r}\right)^2\right) \\
&= \operatorname{mod}(T(s), \gamma_\tau^*) + O'\left(\left(\frac{\Delta s}{\log r}\right)^2\right),
\end{aligned}$$

proving equality (3.6).

From equality (3.1) and inequality (3.5) we have

$$\operatorname{mod}\bigl(T(s), T(t(r))\bigr) \leqslant \operatorname{mod}(T(s), \gamma_t^*).$$

Using (3.1) and (3.6), we also have

$$\operatorname{mod}\bigl(T(s), T(t(r))\bigr) = \operatorname{mod}(T(s), \gamma_t^*) + O'\left(\left(\frac{\Delta s}{\log r}\right)^2\right).$$

Hence there exists $R(r)$, such that

$$\operatorname{mod}\bigl(T(R(r)),T(t(r))\bigr)=\operatorname{mod}(T(s),\gamma_t^*),$$
$$R(r)\leqslant s,\qquad s-R(r)=O'\biggl(\biggl(\frac{\Delta s}{\log r}\biggr)^2\biggr).$$

Now we are ready to define the required function v^* on the set $\overline{(T(R(r)),T)}$. Let f^- be some conformal mapping of the domain $(T(R(r)),\,T(t(r)))$ onto $(T(s),\gamma_t^*)$, and let f^+ be a conformal mapping of the domain $(T(t(r)),\gamma_\tau^*)$ onto $(\gamma_t^*,\gamma_\tau^*)$ such that $f^-(T(t(r)))=f^+(T(t(r)))=\gamma_t^*$. We set

$$v^*(z)=\begin{cases}u^*(f^-(z)), & z\in\overline{(T(R(r)),T(t(r)))},\\ u^*(f^+(z)), & z\in\overline{(T(t(r)),\gamma_\tau^*)},\\ u^*(z), & z\in\overline{(\gamma_\tau^*,T)}.\end{cases}\tag{3.7}$$

Note that the function v^* is locally Lipschitzian on its domain, and hence for it the Dirichlet integral of the form (2.3) is well defined.

To estimate the Dirichlet integral, consider the auxiliary functions

$$\omega_r(z)=\frac{\log s}{\log R(r)}\log|z|\quad\text{and}\quad\sigma_r(z)=\frac{\log t}{\log t(r)}\log|z|.$$

Note that

$$\omega_r(z)=b(z)+O'\biggl(\biggl(\frac{\Delta s}{\log r}\biggr)^2\biggr),\qquad\frac{s}{2}\leqslant|z|\leqslant1,\tag{3.8}$$
$$\sigma_r(z)=b(z)+O'\biggl(\frac{\Delta s}{\log r}\biggr),\qquad\frac{s}{2}\leqslant|z|\leqslant1,\tag{3.9}$$

and

$$b(z_k^*)-\sigma_r(z_k^*)=\frac{c_k}{\log r}+O\biggl(\biggl(\frac{1}{\log r}\biggr)^2\biggr),\tag{3.10}$$

where the constants c_k are equal for the points z_k^* from the circle, $k=1,\dots,mn$.

For sufficiently small $r>0$, consider the sets

$$E(z_k^*,r)$$
$$=\begin{cases}\{z\colon v^*(z)\geqslant\delta_k+b(z_k^*)-\sigma_r(z_k^*)+\sigma_r(z)-(\log r)^{-2},\,|z-z_k^*|\leqslant\sqrt{r}\}, & \delta_k>0,\\ \{z\colon v^*(z)\leqslant\delta_k+b(z_k^*)-\sigma_r(z_k^*)+\sigma_r(z)+(\log r)^{-2},\,|z-z_k^*|\leqslant\sqrt{r}\}, & \delta_k<0,\end{cases}$$

$k=1,\dots,mn$. Note that, for small r, we have $\mathscr{E}(z_k^*,r)\subset E(z_k^*,r)$, $k=1,\dots,mn$. Let $D:=(T(s),T)$. On the boundary of the sets $E(z_k^*,r)$ and $\mathscr{E}(z_k^*,r)$, we have

$$\delta_k+O\biggl(\biggl(\frac{1}{\log r}\biggr)^2\biggr)$$
$$=-\frac{\delta_k}{\log r}\biggl[1+\frac{\log r(D,z_k^*)}{\log r}\biggr][\log r(D,z_k^*)-\log|z-z_k^*|+O(|z-z_k^*|)]$$

$$-\sum_{\substack{j=1\\ j\neq k}}^{mn}\delta_j\left\{\frac{g_D(z_j^*,z_k^*)}{(\log r)^2}[\log r(D,z_k^*)-\log|z-z_k^*|+O(|z-z_k^*|)]+\frac{g_D(z_k^*,z_j^*)}{\log r}\right\}.$$

Hence

$$1=\frac{\log|z-z_k^*|}{\log r}\left\{1+\frac{\log r(D,z_k^*)}{\log r}+\sum_{\substack{j=1\\ j\neq k}}^{mn}\frac{\delta_j}{\delta_k}\,\frac{g_D(z_j^*,z_k^*)}{\log r}\right\}-\frac{\log r(D,z_k^*)}{\log r}$$
$$-\sum_{\substack{j=1\\ j\neq k}}^{mn}\frac{\delta_j}{\delta_k}\,\frac{g_D(z_j^*,z_k^*)}{\log r}+O\left(\left(\frac{1}{\log r}\right)^2\right),$$

and, therefore,

$$1=\frac{\log|z-z_k^*|}{\log r}+O\left(\left(\frac{1}{\log r}\right)^2\right).$$

So, on the sets $\partial E(z_k^*,r)$, $\partial\mathscr{E}(z_k^*,r)$, we have the asymptotic equality

$$|z-z_k^*|=r\left(1+O\left(\frac{1}{\log r}\right)\right),\qquad r\to 0. \tag{3.11}$$

We set $E^*(r)=\bigcup_{k=1}^{mn}E(z_k^*,r)$. Let us now compare the Dirichlet integrals of the form (2.3). Since the Dirichlet integral is conformally invariant, we have, by the Dirichlet principle,

$$I\big(u^*,(T(s),T)\setminus\mathscr{E}^*(r)\big)\geqslant I\big(v^*,(T(R(r)),T)\setminus E^*(r)\big)$$
$$\geqslant I\big(v^*,(T(R(r)),T(t(r)))\big)+I\big(h^*,(T(t(r)),T)\setminus E^*(r)\big).$$

Here, h^* is a harmonic on the set $(T(t(r)),T)\setminus E^*(r)$, continuous in the closure of this set, and is equal to v^* on its boundary. Note that the function h^* is symmetric in the sense of §2. We also note that the circles $T(R(r))$ and T are unchanged under the dissymmetrization Dis from Lemma 2.3. Another appeal to the Dirichlet principle gives us that

$$I\big(h^*,(T(t(r)),T)\setminus E^*(r)\big)=I\big(\operatorname{Dis}h^*,(T(t(r)),T)\setminus E(r)\big)$$
$$\geqslant I\big(h,(T(t(r)),T)\setminus E(r)\big),$$

where $E(r)=\operatorname{Dis}E^*(r)$, and h is harmonic on the set $(T(t(r)),T)\setminus E(r)$, continuous in the closure of this set, and is equal to $\operatorname{Dis}h^*$ on its boundary. Applying again the Dirichlet principle, we find that

$$I\big(v^*,(T(R(r)),T(t(r)))\big)+I\big(h,(T(t(r)),T)\setminus E(r)\big)$$
$$\geqslant I\big(u,(T(R(r)),T)\setminus E(r)\big)$$

where u is harmonic in $(T(R(r)),T)\setminus E(r)$, continuous in the closure of this set, equal to h on $\partial E(r)\cup T$, and equal to $\log s$ on $T(R(r))$. From the above inequalities, we have the inequality

$$I\big(u^*,(T(s),T)\setminus\mathscr{E}^*(r)\big)-I\big(u,(T(R(r)),T)\setminus E(r)\big)$$
$$\geqslant I\big(h^*,(T(t(r)),T)\setminus E^*(r)\big)-I\big(h,(T(t(r)),T)\setminus E(r)\big), \tag{3.12}$$

which is the main ingredient in the proof of Theorem 1.1.

Let us now proceed with estimates of the integrals involved in (3.12). Using the Green formula, and then employing the Gauss theorem and Lemma 2.2 (in view of (3.11)), we have, for the first integral in the left-hand side of (3.12),

$$
\begin{aligned}
&I\bigl(u^*, (T(s),T)\setminus \mathscr{E}^*(r)\bigr)\\
&= I\bigl(b, (T(s),T)\setminus \mathscr{E}^*(r)\bigr) + I\bigl(g_s^*, (T(s),T)\setminus \mathscr{E}^*(r)\bigr) - 2\int_{\partial \mathscr{E}^*(r)} g_s^*\,\frac{\partial b}{\partial n}\,ds\\
&= I\bigl(b, (T(s),T)\setminus \mathscr{E}^*(r)\bigr) - 2\pi\sum_{k=1}^{mn}\frac{\delta_k^2}{\log r}\\
&\quad - 2\pi\biggl\{\sum_{k=1}^{mn}\delta_k^2\log r\bigl((T(s),T), z_k^*\bigr) + E\bigl(Z^*, \Delta, (T(s),T)\bigr)\biggr\}\biggl(\frac{1}{\log r}\biggr)^2 + o\biggl(\biggl(\frac{1}{\log r}\biggr)^2\biggr).
\end{aligned}
\tag{3.13}
$$

To evaluate the second integral on the left of (3.12), we write u as

$$u = \omega_r + g_s,$$

where g_s is harmonic on the set $(T(R(r)),T)\setminus E(r)$, continuous in the closure of this set, vanishes on $T\cup T(R(r))$, and $g_s = h - \omega_r$ on $\partial E(r)$. In view of (3.8) and (3.9), we have

$$
\begin{aligned}
&\int_{\partial E(r)} g_s\,\frac{\partial \omega_r}{\partial n}\,ds\\
&= \sum_{k=1}^{mn}\int_{\partial \operatorname{Dis} E(z_k^*,r)}\biggl[\delta_k + b(z_k) - \sigma_r(z_k) + \sigma_r(z) - \frac{\delta_k}{|\delta_k|}(\log r)^{-2} - \omega_r(z)\biggr]\frac{\partial \omega_r}{\partial n}\,ds\\
&= o\biggl(\biggl(\frac{1}{\log r}\biggr)^2\biggr), \qquad r\to 0.
\end{aligned}
$$

Now the Green formula gives

$$
\begin{aligned}
I\bigl(u, (T(R(r)),T)\setminus E(r)\bigr) &= I\bigl(\omega_r, (T(R(r)),T)\setminus E(r)\bigr)\\
&\quad + I\bigl(g_s, (T(R(r)),T)\setminus E(r)\bigr) + o\biggl(\biggl(\frac{1}{\log r}\biggr)^2\biggr).
\end{aligned}
\tag{3.14}
$$

Next, by definition of the functions ω_r we have

$$
\begin{aligned}
I\bigl(\omega_r, (T(R(r)),T)\setminus E(r)\bigr) &= I\bigl(b, (T(s),T)\setminus E(r)\bigr) + O'\biggl(\biggl(\frac{\Delta s}{\log r}\biggr)^2\biggr)\\
&\geqslant I\bigl(b, (T(s),T)\setminus E(r)\bigr) - d\biggl(\frac{\Delta s}{\log r}\biggr)^2,
\end{aligned}
\tag{3.15}
$$

where d is some positive constant. In the rest of the proof of Theorem 1.1, we add to the above Δs a fixed sufficiently small number $\varepsilon > 0$. For small $r > 0$ and sufficiently large β, on the boundary of the set $\operatorname{Dis} E(z_k^*, r)$ we have $g_s > \delta_k - \beta/(\log r)^2$ if $\delta_k > 0$, and $g_s < \delta_k + \beta/(\log r)^2$ if $\delta_k < 0$ (see (3.8)). Let

a function $\widetilde{g}_s^{\varepsilon}$ be harmonic on the set $(T(s-\varepsilon),T)\setminus E(r)$, continuous in the closure of this set, vanish on $T(s-\varepsilon)\cup T$, and equal to $\delta_k-\beta(\delta_k/|\delta_k|)/(\log r)^2$ on $\partial\operatorname{Dis} E(z_k^*,r)$, $k=1,\dots,mn$. From the Dirichlet principle it easily follows that, for small r,

$$I\big(g_s,(T(R(r)),T)\setminus E(r)\big)\geqslant I\big(\widetilde{g}_s^{\varepsilon},(T(s-\varepsilon),T)\setminus E(r)\big).$$

Substituting this estimate and inequality (3.15) into (3.14) and applying Lemma 2.2 (in view of (3.11)), we have

$$\begin{aligned}I\big(u,(T(R(r)),T)\setminus E(r)\big)&\geqslant I\big(b,(T(s),T)\setminus E(r)\big)\\&-2\pi\sum_{k=1}^{mn}\frac{\delta_k^2}{\log r}-2\pi\Bigg\{\sum_{k=1}^{mn}\delta_k^2\log r\big((T(s-\varepsilon),T),z_k\big)\\&+E\big(Z,\Delta,(T(s-\varepsilon),T)\big)+\frac{d}{2\pi}(\Delta s)^2\Bigg\}\left(\frac{1}{\log r}\right)^2+o\left(\left(\frac{1}{\log r}\right)^2\right).\end{aligned}\tag{3.16}$$

To evaluate the first integral on the right of (3.12), we write the function h^* as

$$h^*=\sigma_r+g_t^*,$$

where g_t^* is harmonic в $(T(t(r)),T)\setminus E^*(r)$, continuous in the closure of this set, vanishes on $T(t(r))\cup T$, and

$$g_t^*=\delta_k+b(z_k^*)-\sigma_r(z_k^*)-\frac{\delta_k/|\delta_k|}{(\log r)^2}=\delta_k+\frac{c_k}{\log r}+O\left(\left(\frac{1}{\log r}\right)^2\right)$$

on $\partial E(z_k^*,r)$, $k=1,\dots,mn$ (see (3.10)). Applying again the Green formula and the Gauss theorem, we have

$$\begin{aligned}I\big(h^*,(T(t(r)),T)\setminus E^*(r)\big)&=I\big(\sigma_r,(T(t(r)),T)\setminus E^*(r)\big)\\&+I\big(g_t^*,(T(t(r)),T)\setminus E^*(r)\big).\end{aligned}\tag{3.17}$$

Let r be so small that the domain $(T(t-\varepsilon),T)$ contains $(T(t(r)),T)$, and let a function $g_t^{*\varepsilon}$ be harmonic in $(T(t-\varepsilon),T)\setminus E^*(r)$, continuous in the closure of this set, vanish on $T(t-\varepsilon)\cup T$, and $g_t^{*\varepsilon}=g_t^*$ on $\partial E(z_k^*,r)$, $k=1,\dots,mn$. By the Dirichlet principle and Lemma 2.2 (in view of (3.10) and (3.11)) we have

$$\begin{aligned}I\big(g_t^*,(T(t(r)),T)\setminus E^*(r)\big)&\geqslant I\big(g_t^{*\varepsilon},(T(t-\varepsilon),T)\setminus E^*(r)\big)\\&=-2\pi\sum_{k=1}^{mn}\frac{\delta_k^2}{\log r}-2\pi\Bigg\{\sum_{k=1}^{mn}\big[2c_k\delta_k+\delta_k^2\log r\big((T(t-\varepsilon),T),z_k^*\big)\big]\\&+E\big(Z^*,\Delta,(T(t-\varepsilon),T)\big)\Bigg\}\left(\frac{1}{\log r}\right)^2+o\left(\left(\frac{1}{\log r}\right)^2\right).\end{aligned}\tag{3.18}$$

To estimate from above the second integral on the right of (3.12), we write g as

$$h=\sigma_r+g_t,$$

where the function g_t is a harmonic in $(T(t(r)),T)\setminus E(r)$, continuous in the closure of this set, is zero on $T(t(r))\cup T$, and $g_t=\delta_k+b(z_k)-\sigma_r(z_k)-(\delta_k/|\delta_k|)/(\log r)^2$

on $\partial \operatorname{Dis} E(z_k^*, r)$, $k = 1, \dots, mn$. Let, as above, $\varepsilon > 0$, and let r be such that the domain $(T(t+\varepsilon), T)$ is contained in $(T(t(r)), T)$. We let g_t^ε denote a function which is harmonic on the set $(T(t+\varepsilon), T) \setminus E(r)$, continuous in the closure of this set, vanishes on $T(t+\varepsilon) \cup T$, and $g_t^\varepsilon = g_t$ on $\partial \operatorname{Dis} E(z_k^*, r)$, $k = 1, \dots, mn$. By the Dirichlet principle,

$$I\big(g_t, (T(t(r)), T) \setminus E(r)\big) \leqslant I\big(g_t^\varepsilon, (T(t+\varepsilon), T) \setminus E(r)\big).$$

Using again the Green formula and Lemma 2.2 (in view of (3.10) and (3.11)), we find that

$$\begin{aligned}
&I\big(h, (T(t(r)), T) \setminus E(r)\big) \\
&\quad = I\big(\sigma_r, (T(t(r)), T) \setminus E(r)\big) + I\big(g_t, (T(t(r)), T) \setminus E(r)\big) \\
&\quad \leqslant I\big(\sigma_r, (T(t(r)), T) \setminus E(r)\big) - 2\pi \sum_{k=1}^{mn} \frac{\delta_k^2}{\log r} \\
&\qquad - 2\pi \Bigg\{ \sum_{k=1}^{mn} \big[2c_k\delta_k + \delta_k^2 \log r\big((T(t+\varepsilon), T), z_k\big)\big] \\
&\qquad\qquad + E\big(Z, \Delta, (T(t+\varepsilon), T)\big) \Bigg\} \left(\frac{1}{\log r}\right)^2 + o\left(\left(\frac{1}{\log r}\right)^2\right). \qquad (3.19)
\end{aligned}$$

Substituting (3.13), (3.16)–(3.19) into (3.12) and cancelling, we have, as $r \to 0$,

$$\begin{aligned}
&\frac{d}{2\pi}(\Delta s)^2 - \sum_{k=1}^{mn} \delta_k^2 \log r((T(s), T), z_k^*) - E\big(Z^*, \Delta, (T(s), T)\big) \\
&\qquad + \sum_{k=1}^{mn} \delta_k^2 \log r\big((T(s-\varepsilon), T), z_k\big) + E\big(Z, \Delta, (T(s-\varepsilon), T)\big) \\
&\quad \geqslant - \sum_{k=1}^{mn} \delta_k^2 \log r\big((T(t-\varepsilon), T), z_k^*\big) - E\big(Z^*, \Delta, (T(t-\varepsilon), T)\big) \\
&\qquad + \sum_{k=1}^{mn} \delta_k^2 \log r\big((T(t+\varepsilon), T), z_k\big) + E\big(Z, \Delta, (T(t+\varepsilon), T)\big).
\end{aligned}$$

The convergence, as $\varepsilon \to 0$, of the Green functions, and, therefore, of the inner radii of the domains to the corresponding Green functions and the inner radii of the limit domains follows, for example, from the Hadamard variational formula for Green functions (see [13], A3.3) and Harnack's theorem for harmonic functions. From the resulting inequality we get a differential inequality for the Green energies, which, in turn, secures (1.1). Theorem 1.1 is proved.

§ 4. Inequalities for complex numbers

We first consider a particular case of Theorem 1.1 mentioned in the introduction.

Corollary 4.1. *Let $0<\rho<t<1$, let z_k, $k=1,\dots,n$, be arbitrary points on the circle $|z|=\rho$, and $z_k^*=\rho\exp(2\pi ik/n)$, $k=1,\dots,n$, be symmetric points on this circle. Then*

$$\prod_{k=1}^{n}\prod_{l=1}^{n}\left|\frac{1-z_k\overline{z}_l}{t^2-z_k\overline{z}_l}\right|\geqslant\prod_{k=1}^{n}\prod_{l=1}^{n}\left|\frac{1-z_k^*\overline{z_l^*}}{t^2-z_k^*\overline{z_l^*}}\right|. \tag{4.1}$$

Proof. It can be assumed that, for some values of the arguments,

$$\arg z_1<\arg z_2<\dots<\arg z_n<\arg z_1+2\pi.$$

Applying Theorem 1.1 to $Z=\{z_k\}_{k=1}^n$, $\Delta=\{1,\dots,1\}$ and $s_1=t_1=0$, $s_2=1$, $t_2=t$, $\rho_1=\rho$ $(m=1)$, we get

$$\sum_{k=1}^{n}\sum_{\substack{l=1\\l\neq k}}^{n}\log\left|\frac{1-z_k\overline{z}_l}{z_k-z_l}\right|-\sum_{k=1}^{n}\sum_{\substack{l=1\\l\neq k}}^{n}\log\left|\frac{1-z_k^*\overline{z_l^*}}{z_k^*-z_l^*}\right|$$
$$\geqslant\sum_{k=1}^{n}\sum_{\substack{l=1\\l\neq k}}^{n}\log\left|\frac{t^2-z_k\overline{z}_l}{t(z_k-z_l)}\right|-\sum_{k=1}^{n}\sum_{\substack{l=1\\l\neq k}}^{n}\log\left|\frac{t^2-z_k^*\overline{z_l^*}}{t(z_k^*-z_l^*)}\right|.$$

This proves inequality (4.1), and, therefore, Corollary 4.1.

Given fixed numbers θ_k, $k=1,\dots,n$, $n\geqslant 2$, $\theta_1<\theta_2<\dots<\theta_n<\theta_1+2\pi$, and a fixed ρ, $1<\rho<\infty$, we set

$$f(\rho)=\frac{\prod_{k=1}^{n}\prod_{l=1}^{n}|z_k-\zeta_l|}{\prod_{k=1}^{n}\prod_{l=1}^{n}|z_k^*-\zeta_l^*|},$$

where $z_k=\exp(i\theta_k)$, $\zeta_k=\rho\exp(i\theta_k)$, $z_k^*=\exp(2\pi ik/n)$, $\zeta_k^*=\rho\exp(2\pi ik/n)$, $k=1,\dots,n$. We also set

$$f(1):=\lim_{\rho\to 1}f(\rho)=\frac{\prod_{k=1}^{n}\prod_{\substack{l=1\\l\neq k}}^{n}|z_k-z_l|}{\prod_{k=1}^{n}\prod_{\substack{l=1\\l\neq k}}^{n}|z_k^*-z_l^*|}.$$

Corollary 4.2. *The function $f(\rho)$ is non-decreasing on $1\leqslant\rho<\infty$.*

Proof. Let $1<\rho_1<\rho_2$. We rewrite inequality (4.1) for $\rho=\sqrt{1/\rho_2}$, $t=\sqrt{\rho_1/\rho_2}$ and $z_k\mapsto z_k\rho$ as

$$\prod_{k=1}^{n}\prod_{l=1}^{n}\left|\frac{1-z_k\overline{z}_l\rho^2}{1-z_k^*\overline{z_l^*}\rho^2}\right|\geqslant\prod_{k=1}^{n}\prod_{l=1}^{n}\left|\frac{t^2-z_k\overline{z}_l\rho^2}{t^2-z_k^*\overline{z_l^*}\rho^2}\right|. \tag{4.2}$$

The left-hand side of (4.2) is

$$\prod_{k=1}^{n}\prod_{l=1}^{n}\left|\frac{z_k-\rho_2/\overline{z}_l}{z_k^*-\rho_2/\overline{z_l^*}}\right|=f(\rho_2).$$

Similarly, the right-hand side of (4.2) is

$$\prod_{k=1}^{n}\prod_{l=1}^{n}\left|\frac{z_k-t^2\rho_2/\overline{z}_l}{z_k^*-t^2\rho_2/\overline{z_l^*}}\right|=\prod_{k=1}^{n}\prod_{l=1}^{n}\left|\frac{z_k-\rho_1/\overline{z}_l}{z_k^*-\rho_1/\overline{z_l^*}}\right|=f(\rho_1).$$

Now applying (4.2) we get $f(\rho_2)\geqslant f(\rho_1)$, proving Corollary 4.2.

Corollary 4.3. *For all points z_k, $k = 1, \dots, n$, on the circle $|z| = 1$, and any number $\rho > 1$,*

$$\prod_{k=1}^{n} \prod_{l=1}^{n} |z_k - \zeta_l| \leqslant \prod_{k=1}^{n} \prod_{l=1}^{n} |z_k^* - \zeta_l^*| = (\rho^n - 1)^n, \tag{4.3}$$

where $\zeta_k = z_k \rho$, $\zeta_k^* = z_k^* \rho$ and $z_k^* = \exp(2\pi i k/n)$, $k = 1, \dots, n$.

Proof. The inequality on the left of (4.3) follows from Corollary 4.2, because $f(\rho) \to 1$ as $\rho \to \infty$. The equality on the right of (4.3) can be verified directly or by comparing the capacity of an appropriate n-fold symmetric condenser with that of its range under the power function $w = z^n$ (see [8], § 2.15). This proves Corollary 4.3.

Inequality (4.3) is a natural generalization of the classical Pólya–Schur inequality (see, for example, [15], § 3):

$$\prod_{k=1}^{n} \prod_{\substack{l=1 \\ l \neq k}}^{n} |z_k - z_l| \leqslant \prod_{k=1}^{n} \prod_{\substack{l=1 \\ l \neq k}}^{n} |z_k^* - z_l^*| = n^n. \tag{4.4}$$

Inequality (4.4), which is equivalent to the inequality $f(1) \leqslant 1$ for the function f from Corollary 4.2, can also be derived from (4.3) by dividing both parts of (4.3) by $(\rho - 1)^n$ and then making $\rho \to 1$. For other generalizations and extensions of inequality (4.4) obtained via the machinery of condensers, see [8], § 5.1. Among the known strengthenings of the Pólya–Schur inequality, we mention the following result of Fejes Tóth [16], who in 1956 proved that the sum

$$\sum_{k=1}^{n} \sum_{l=1}^{n} |z_k - z_l|$$

assumes its maximum value on the circle if the points (z_k) are symmetric. There is an elementary proof of inequality (4.4), but the author of the present paper is unaware of any simple proof of inequality (4.3), not to speak of Corollary 4.2. The following observation shows that, for $\rho > 1$, the proof of inequality (4.3) is much more involved than that of (4.4). Indeed, it is easily checked that, for $\rho > 1$ and sufficiently large n,

$$|z_1^* - \zeta_2^*|\,|\zeta_2^* - z_3^*| < |z_1^* - \zeta_2|\,|\zeta_2 - z_3^*|$$

for any point ζ_2 lying on the circle $|z| = \rho$ strictly between ζ_1^* and ζ_3^*. At the same time, for $\rho = 1$ ($\zeta_k^* = z_k^*$, $k = 1, 2, 3$), the opposite inequality also holds.

Corollary 4.4. *Let $0 < \rho_1 < \rho_2 < 1$, z_k, $k = 1, \dots, n$, be arbitrary points on the circle $|z| = \rho_1$, and let ζ_k, $k = 1, \dots, n$, be points on the circle $|z| = \rho_2$ satisfying $\arg \zeta_k = \arg z_k$, $k = 1, \dots, n$. Then*

$$\prod_{k=1}^{n} \prod_{l=1}^{n} \left| \frac{z_k - \zeta_l}{1 - z_k \overline{\zeta}_l} \right| \leqslant \prod_{k=1}^{n} \prod_{l=1}^{n} \left| \frac{z_k^* - \zeta_l^*}{1 - z_k^* \overline{\zeta}_l^*} \right|, \tag{4.5}$$

where $z_k^* = \rho_1 \exp(2\pi i k/n)$, $\zeta_k^* = \rho_2 \exp(2\pi i k/n)$, $k = 1, \dots, n$.

Proof. We define the function $f(\rho)$ for $\theta_k = \arg z_k$, $k = 1, \dots, n$, $\rho > 1$, and apply the inequality

$$f\left(\frac{\rho_2}{\rho_1}\right) \leqslant f\left(\frac{1}{\rho_2\rho_1}\right),$$

which follows from Corollary 4.2. As a result, we get the inequality

$$\frac{\prod_{k=1}^n \prod_{l=1}^n |z_k/\rho_1 - (\zeta_l/\rho_2)(\rho_2/\rho_1)|}{\prod_{k=1}^n \prod_{l=1}^n |z_k^*/\rho_1 - (\zeta_l^*/\rho_2)(\rho_2/\rho_1)|} \leqslant \frac{\prod_{k=1}^n \prod_{l=1}^n |z_k/\rho_1 - (\zeta_l/\rho_2)(1/(\rho_1\rho_2))|}{\prod_{k=1}^n \prod_{l=1}^n |z_k^*/\rho_1 - (\zeta_l^*/\rho_2)(1/(\rho_1\rho_2))|}.$$

Since $\zeta_l/\rho_2^2 = 1/\overline{\zeta}_l$, $l = 1, \dots, n$, we arrive at (4.5), proving Corollary 4.4.

Making $\rho_2 \to \rho_1$, in (4.5), we get the inequality (see [17], Corollary 3)

$$\prod_{k=1}^n \prod_{\substack{l=1 \\ l\neq k}}^n \left|\frac{z_k - z_l}{1 - z_k\overline{z}_l}\right| \leqslant \prod_{k=1}^n \prod_{\substack{l=1 \\ l\neq k}}^n \left|\frac{z_k^* - z_l^*}{1 - z_k^*\overline{z_l^*}}\right|. \tag{4.6}$$

Note that by (4.3) we have

$$\prod_{k=1}^n \prod_{\substack{l=1 \\ l\neq k}}^n |1 - z_k\overline{z}_l| \leqslant \prod_{k=1}^n \prod_{\substack{l=1 \\ l\neq k}}^n |1 - z_k^*\overline{z_l^*}|.$$

So, (4.6) strengthens the Pólya–Schur inequality (4.4).

Inequality (4.3) means that the mutual logarithmic energy

$$E_{\log}(\{z_k\}_{k=1}^n, \{\zeta_k\}_{k=1}^n) := -\sum_{k=1}^n \sum_{l=1}^n \log|z_k - \zeta_l|$$

of two discrete measures, of which one is concentrated at points of the circle $|z| = 1$, and the other one, on the circle $|z| = \rho$, assumes its smallest value if the points are symmetric. In a similar way, inequality (4.5) says that the mutual Green energy relative to the disc $|z| < 1$ of two discrete measures on the circles $|z| = \rho_1$ and $|z| = \rho_2$ decreases (rather than increases) when changing from arbitrary points to symmetric ones.

§ 5. Counterexamples and open problems

We start with two assumptions about the behaviour of the mutual energy of measures under symmetrization. According to the above, the mutual logarithmic energy and the mutual Green energy relative to the disc $|z| < 1$ do not increase when changing to symmetric measures. It can be anticipated, therefore, that, in the case of the Green energy, the disc $|z| < 1$ in this result can be replaced by a more involved domain. Let us show that this is not the case even for a disc with radial cuts, and for a considerably simplified variant of the problem. Let r and θ be fixed numbers, $0 < r < 1$, $0 < \theta < \pi$. We set:

$$E = \{z\colon r \leqslant |z| \leqslant 1, \arg z = 0 \text{ or } \arg z = \theta\},$$
$$E^* = \{z\colon r \leqslant |z| \leqslant 1, \arg z = 0 \text{ or } \arg z = \pi\},$$
$$U = \{z\colon |z| < 1\}, \qquad B = U \setminus E, \qquad B^* = U \setminus E^*.$$

For ρ, $r < \rho < 1$, and sufficiently small $\varepsilon > 0$, consider two discrete measures, of which one, say, of size 1, is concentrated at the origin, and the other one assumes the same value 1 at the four points $z_1 = \rho e^{i\varepsilon}$, $z_2 = \rho e^{-i\varepsilon}$, $z_3 = \rho e^{i(\theta+\varepsilon)}$, $z_4 = \rho e^{i(\theta-\varepsilon)}$. The mutual Green energy of these two measures relative to the domain B is

$$\sum_{k=1}^{4} g_B(0, z_k).$$

Similarly, we consider the *symmetric* pair of discrete measures with mutual energy

$$\sum_{k=1}^{4} g_{B^*}(0, z_k^*),$$

where $z_1^* = z_1$, $z_2^* = z_2$, $z_3^* = -\rho e^{i\varepsilon}$, $z_4^* = -\rho e^{-i\varepsilon}$. Assume that, for all ρ, $r < \rho < 1$, and all sufficiently small $\varepsilon > 0$, the mutual energy of the symmetric measures is majorized by that of the first two measures. Then, for any ρ, $r < \rho < 1$, on the intersection of the circle $|z| = \rho$ with E^*, the total density of the harmonic measure relative to the domain B^* and the point $z = 0$ is not smaller than that of the harmonic measure relative to the domain B and $z = 0$. Therefore, the harmonic measure of the set E^* relative to the domain B^* (as calculated at 0) is not greater than the harmonic measure of E relative to B at 0. But this contradicts the solution of the Gonchar problem on harmonic measure (see [18], [8], Theorem 4.17). Note that, unlike the mutual energy, the Green energy relative to the domain B of the measure concentrated at the points $\{z_k\}_{k=1}^4$ is not smaller than the Green energy relative to B^* of the measure at the points $\{z_k^*\}_{k=1}^4$ for all ρ, $r < \rho < 1$, and small $\varepsilon > 0$ (see [8], Theorem 4.15).

Let us return back to the Green energy relative to the disc U. Setting $\rho_1 = \rho$ in (4.5) and making $\rho_2 \to \rho$, we conclude that the logarithmic energy of the discrete measure concentrated on the circle $|z| = \rho$ minus the mutual energy of this measure and the measure on the circle $|z| = 1/\rho$ does not increase when changing to the symmetric case. It is natural to expect here that the last fact should also hold for the circles $|z| = \rho_k$, $k = 1, 2$, $0 < \rho_1 < \rho_2 < 1$, if the logarithmic energy is replaced by the Green energy relative to the disc U. However this is not the case. Indeed, let us compare this difference in the arbitrary and symmetric cases for two quadruples of points $z_1 = \rho_1 e^{i\theta}$, $z_2 = \rho_1$, $\zeta_1 = \rho e^{i\theta}$, $\zeta_2 = \rho$ ($\rho \equiv \rho_2 > \rho_1$, $0 < \theta < \pi$), $z_1^* = -\rho_1$, $z_2^* = \rho_1$, $\zeta_1^* = -\rho$, $\zeta_2^* = \rho$. We have

$$\begin{aligned} &2g_U(\zeta_1, \zeta_2) - [g_U(z_1, \zeta_1) + g_U(z_1, \zeta_2) + g_U(z_2, \zeta_1) + g_U(z_2, \zeta_2)] \\ &\qquad - 2g_U(\zeta_1^*, \zeta_2^*) + [g_U(z_1^*, \zeta_1^*) + g_U(z_1^*, \zeta_2^*) + g_U(z_2^*, \zeta_1^*) + g_U(z_2^*, \zeta_2^*)] \\ &\quad = 2[g_U(\zeta_1, \zeta_2) - g_U(\zeta_1^*, \zeta_2^*)] - 2[g_U(z_2, \zeta_1) - g_U(z_2^*, \zeta_1^*)]. \end{aligned}$$

Each term in the left square parentheses behaves as $O((1-\rho)^2)$ as $\rho \to 1$. For example,

$$\begin{aligned} g_U(\zeta_1, \zeta_2) &= \log\left|\frac{1 - \zeta_1\overline{\zeta}_2}{\zeta_1 - \zeta_2}\right| = \log\left|\frac{1 - \rho^2 e^{i\theta}}{\rho(1 - e^{i\theta})}\right| \\ &= \log\left|\frac{2(1-\rho)e^{i\theta}/(1 - e^{i\theta}) + O((1-\rho)^2)}{1 - (1-\rho)}\right| \end{aligned}$$

$$= (1-\rho)\operatorname{Re}\frac{1+e^{i\theta}}{1-e^{i\theta}} + O((1-\rho)^2) = O((1-\rho)^2).$$

A similar analysis shows that the difference in the right square parentheses is of order $(1-\rho)$. So, as $\rho \to 1$, the sign of the difference is controlled by the quantity inside the parentheses on the right. On the other hand, the strict inequality

$$g_U(z_2, \zeta_1) > g_U(z_2^*, \zeta_1^*),$$

can either be verified directly or from the well-known property of polarization relative to the ray $\arg z = (\pi - \theta)/2$ (see [8], Theorem 3.6, formula (2.16)). Hence the difference under consideration is negative if ρ is close to 1, which contradicts the assumption.

We mention the following open problems.

1. Let $B = \{z\colon s < |z| < t\}$, $0 < s < \rho_1 < \rho_2 < t < \infty$; z_k, $k = 1, \dots, n$, be arbitrary point on the circle $|z| = \rho_1$, $\zeta_k = \rho_2 z_k/|z_k|$, $k = 1, \dots, n$. Prove that mutual energy

$$\sum_{k=1}^{n}\sum_{l=1}^{n} g_B(z_k, \zeta_l) \tag{5.1}$$

is smallest in the case of symmetric points $z_k = \rho_1 \exp(2\pi i k/n)$, $k = 1, \dots, n$.

2. Let the domain B be as above; z_k, $k = 1, \dots, n$, be arbitrary points on the circle $|z| = \rho_1$, and ζ_k, $k = 1, \dots, n$, be arbitrary points on the circle $|z| = \rho_2$. Is it true that the mutual energy (5.1) is smallest in the case $z_k = \rho_1 \exp(2\pi i k/n)$, $\zeta_k = \rho_2 \exp(i(\pi + 2\pi k)/n)$, $k = 1, \dots, n$?

3. Under the conditions of Problem 2, what can be said about the minimum of the mutual logarithmic energy $E_{\log}(\{z_k\}_{k=1}^n, \{\zeta_k\}_{k=1}^n)$?

4. Under the conditions of Problem 2, find the smallest Green energy relative to the annulus B of the signed measure which is equal to 1 at the points z_k, $k = 1, \dots, n$, and equal to -1 at the points ζ_k, $k = 1, \dots, n$, or, alternatively, which is equal to 1 at all points z_k and ζ_k, $k = 1, \dots, n$.

5. Find the smallest Green energy relative to the annulus B of the signed measure that has the same value at arbitrary n points on each of the circles $|z| = \rho_k$, $k = 1, \dots, m$, $m \geqslant 3$, $0 \leqslant s < \rho_1 < \rho_2 < \dots < \rho_m < t \leqslant \infty$.

6. Consider the same problems for energies generated by the Riesz kernels. For example, prove an analogue of inequality (4.3).

Bibliography

[1] J. S. Brauchart, "Optimal logarithmic energy points on the unit sphere", *Math. Comp.* **77**:263 (2008), 1599–1613.

[2] J. S. Brauchart, D. P. Hardin, and E. B. Saff, "The Riesz energy of the Nth roots of unity: an asymptotic expansion for large N", *Bull. Lond. Math. Soc.* **41**:4 (2009), 621–633.

[3] J. S. Brauchart, D. P. Hardin, and E. B. Saff, "The next-order term for optimal Riesz and logarithmic energy asymptotics on the sphere", *Recent advances in orthogonal polynomials, special functions, and their applications*, Contemp. Math., vol. 578, Amer. Math. Soc., Providence, RI 2012, pp. 31–61.

[4] D. P. Hardin, A. P. Kendall, and E. B. Saff, "Polarization optimality of equally spaced points on the circle for discrete potentials", *Discrete Comput. Geom.* **50**:1 (2013), 236–243.

[5] S. V. Borodachov, D. P. Hardin, A. Reznikov, and E. B. Saff, "Optimal discrete measures for Riesz potentials", *Trans. Amer. Math. Soc.* **370**:10 (2018), 6973–6993.

[6] S. V. Borodachov, D. P. Hardin, and E. B. Saff, *Discrete energy on rectifiable sets*, Springer Monogr. Math., Springer, New York 2019.

[7] N. S. Landkof, *Foundations of modern potential theory*, Nauka, Moscow 1966; English transl., Grundlehren Math. Wiss., vol. 180, Springer-Verlag, New York–Heidelberg 1972.

[8] V. N. Dubinin, *Condenser capacities and symmetrization in geometric function theory*, Springer, Basel 2014.

[9] V. N. Dubinin, "Green energy and extremal decompositions", *Probl. Anal. Issues Anal.* **8(26)**:3 (2019), 38–44.

[10] V. N. Dubinin, "Pólya–Schur inequality and the Green energy of a discrete charge", *Dokl. Akad. Nauk. Matem., inform., proc. upr.* **492** (2020), 24–26; English transl., *Dokl. Math.* **101**:3 (2020), 192–194.

[11] N. I. Akhiezer, *Elements of the theory of elliptic functions*, 2nd ed., Nauka, Moscow 1970; English transl., Transl. Math. Monogr., vol. 79, Amer. Math. Soc., Providence, RI 1990.

[12] V. N. Dubinin, "Asymptotics for the capacity of a condenser with variable potential levels", *Sibirsk. Mat. Zh.* **61**:4 (2020), 796–802; English transl., *Siberian Math. J.* **61**:4 (2020), 626–631.

[13] M. Schiffer, "Some recent developments in the theory of conformal mapping", Appendix to: R. Courant, *Dirichlet's principle, conformal mapping, and minimal surfaces*, Interscience Publishers, Inc., New York, N.Y. 1950, pp. 249–323; Russian transl., Inostr. Lit., Moscow 1953, pp. 234–301.

[14] S. P. Suetin, "An analogue of the Hadamard and Schiffer variational formulas", *Teor. Mat. Fiz.* **170**:3 (2012), 335–341; English transl., *Theoret. and Math. Phys.* **170**:3 (2012), 274–279.

[15] I. Schur, "Über die Verteilung der Wurzeln bei gewissen algebraischen Gleichungen mit ganzzahligen Koeffizienten", *Math. Z.* **1**:4 (1918), 377–402.

[16] L. Fejes Tóth, "On the sum of distances determined by a pointset", *Acta Math. Acad. Sci. Hungar.* **7** (1956), 397–401.

[17] V. N. Dubinin, "Some properties of the reduced inner modulus", *Sibirsk. Mat. Zh.* **35**:4 (1994), 774–792; English transl., *Siberian Math. J.* **35**:4 (1994), 689–705.

[18] V. N. Dubinin, "On the change in harmonic measure under symmetrization", *Mat. Sb.* **124(166)**:2(6) (1984), 272–279; English transl., *Sb. Math.* **52**:1 (1985), 267–273.

Vladimir N. Dubinin
Institute of Applied Mathematics,
Far-Eastern Branch
of the Russian Academy of Sciences, Vladivostok
E-mail: dubinin@iam.dvo.ru,
vdubinin51@gmail.com

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